Raymond Y. Sheng
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Education:

Cornell University, College of Engineering Ithaca, NY
Bachelor of Science, Electrical and Computer Engineering | GPA: 3.973/4.300 Expected May 2028
Relevant Courses: Digital Logic & Computer Organization; Embedded Systems; Signals & Systems; Computer Systems Programming; Circuits for Electrical & Computer Engineers; Probability & Inference for Random Signals & Systems

Work Experience:

General Atomics Aeronautical Systems, Inc. **June 2026 - August 2026**
Systems Engineering Intern San Diego, CA
- Created a software integration prototype for General Atomics' Gray Eagle UAS and U.S. Army's Next-Generation Command and Control initiative, sparking GA's initial investment in NGC2 and driving adoption company-wide.
- Reimaged a Ground Control Station, installing flight monitoring and test software used by 15+ engineers.
- Designed 50+ requirements and 20+ verification test cases for US Army Next-Gen C2 technology.

Cornell University Physical Intelligence **January 2026 - Present**
Electrical Engineer Ithaca, NY
- Designed a 4-layer power-distribution PCB for an 18-motor hexapod, integrating a 48 V-class battery bus with transient, surge, and reverse polarity protection with 12 V, 5 V, and 3.3 V supply rails.
- Designed custom buck converter and LDO boards in Altium to support testing and integration into hexapod robot.
- Built the team as a founding member through early project development and member recruitment.

Cornell Hyperloop **November 2025 - Present**
Power Systems Electrical Engineer Ithaca, NY
- Developed a custom Battery Management System and battery for a 12V 300Ah pack, designing active cell balancing circuitry for cell voltage and monitoring temperature and state of charge to ensure safe operation.
- Designed custom EPO (Emergency Power Off) circuit schematic and PCB layout validating 8+ power regulation, protection, and control circuits in SPICE to catch design flaws before fabrication.

Projects:

Hexapod Power-Distribution Board
- Designed a 4-layer PCB in Altium for a 13S LiPo battery (54.6 V full charge), with a nominal 20 A current-limit target, reverse-polarity protection, and surge-control circuitry.
- Developed a multi-rail power architecture for an 18-motor hexapod, integrating 12 V, 5 V, and 3.3 V buck converters to supply control electronics alongside the motor power bus.

8-Bit CPU
- Architected a custom 8-bit instruction set (8 opcodes, 4 registers) and single-cycle datapath from scratch, balancing register width against instruction encoding under an 8-bit word constraint.
- Implemented and unit-verified 5 RTL modules: register file, ALU, control unit, and instruction/data memory with Icarus Verilog testbenches prior to full datapath integration.

Emergency Power Off Board
- Designed a safety-critical 2-layer PCB in KiCAD with a latching, dual-relay emergency cutoff stage driven by 3 logic-level MOSFETs, protected by flyback diodes and fuse-based overcurrent safeguards.
- Verified power cutoff within 50ms of trigger activation through SPICE simulation, ensuring reliable shutdown for the Hyperloop pod.

Bluetooth Speaker PCB
- Designed a custom 2-layer Bluetooth speaker using stereo Class-D amplification rated for up to 3 W per channel.
- Engineered 5 V power distribution and decoupling networks to support two independent audio channels.
- Created custom Bluetooth module footprints and optimized speaker connections to minimize electrical noise.

Technical Skills:

Design Tools: Altium, KiCAD, Solidworks, TinkerCAD, Autodesk Inventor, LTspice
Hardware: Power Electronics, Analog Circuit Design, RISC-V, FPGA, Raspberry Pi, ESP32, UART, RF Measurement
Software: Verilog, Claude, Cursor, Docker, Python, Java, C++, C, Linux (Red Hat, Ubuntu), Bash, MATLAB, R, Git
Interests: Badminton, Video Game Development, Soccer, Chess, Track and Field, Poker, Board Games